ARCHITECTURE OVERVIEW

Orion

An AI Business Operating System — five departments that generate a company's opening state, then operate it continuously under human supervision, without optimising themselves into failure.

Vinayak Maurya

Department of Computer Engineering · University of Mumbai

27 July 2026

This document describes the Orion architecture as specified. Figures marked measured report data from the evaluation harness; figures marked prospective illustrate a design intent that has not yet been measured.

1 Executive Summary

Orion is an operating system for a company. Five departments — Strategy, Finance, Branding, Marketing and Operations, staffed by twenty-one specialised agents — generate a complete founding plan from a single sentence, and then keep operating the business that plan describes.

The system has two layers. The **generation layer** turns a natural-language prompt into a founding plan: market positioning, unit economics, brand, customer acquisition and day-to-day operations, grounded where possible in retrieved real-world data. The **operating layer** treats that plan not as the deliverable but as the company's opening state, and runs the same five departments on cadences thereafter — planning, executing, remembering and adapting, with humans supervising the decisions that matter.

Building the generation layer produced two findings that shape everything above it. First, cross-agent constraints expressed as prompt instructions were ignored repeatedly, even when they contained explicit numeric bounds; constraints that matter have to be enforced in code. Second, decomposing a quantitative estimate across dependent agents compounds variance: on financial accuracy the pipeline's best run beat a single-LLM baseline's best, and its spread was 7.6× against the baseline's 1.8×. It is unreliably better rather than reliably worse.

Both findings say the same thing in different registers. **An agent cannot be trusted to assert a quantity it is judged on.** The generation layer's answer was local: where two agents assert the same value, the derived figure wins. The operating layer generalises it. Every number by which the business judges itself is computed by code from an append-only record, and no agent has a write path to one. Departments may propose anything; what they cannot do is author the measure of their own success.

> **IN ONE SENTENCE**
> Orion generates a company and then operates it, built so that the parts which check the work cannot be captured by the parts doing it.

2 The Idea in Plain Language

An entrepreneur opening a small business needs a plan covering positioning, unit economics, brand, acquisition and operations. Professional consulting for this costs a great deal. Templates give structure without analysis. General-purpose prompting produces text that reads well and does not survive arithmetic — a revenue projection above the market size the same document just calculated.

Orion's generation layer addresses that by giving each analytical concern its own agent and forcing all of them through one shared state object, so that disagreements between them become visible instead of being silently resolved by whichever agent wrote last.

But a plan is a photograph. The business it describes changes the day after it is written. So the operating layer keeps the same five departments running: the plan becomes the configuration, not the output, and the company acquires a state that persists — customers, spend, decisions, and what it learned last month and why it stopped doing the thing it was doing before.

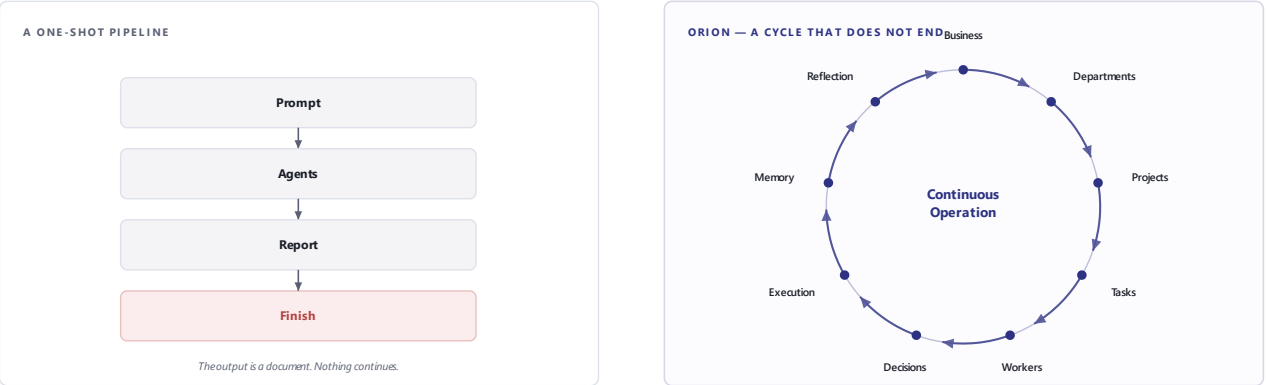


Figure 1 — A one-shot pipeline terminates by design: its output is a document. Orion's cycle has no terminal state — work produces outcomes, outcomes become record, record informs the next plan.

3 The Generation Layer

A natural-language prompt passes through a hybrid intent classifier — the model extracts business type, sector and location hints, while deterministic code parses budget and normalises locality. The split is deliberate: every financial projection is anchored to the budget, so if budget parsing varied by provider, variance in the output could no longer be attributed to the agents.

An orchestrator then decides which departments are active and compiles an execution graph. Strategy runs first, because Finance prices against its competitor set and Marketing targets its personas. Finance, Branding and Marketing then run concurrently. Operations joins on all three, and is skipped entirely for a business with no physical premises — which is a configuration flag rather than a separate code path, so both arms of the routing experiment run identical code.

Two mechanisms in this layer matter more than the topology, and they are the ones the operating layer inherits.

The append-only state contract

All agents read and write one shared state object, and an agent may write only its own department's section. A cross-department write raises a contract violation. This is what makes cross-department agreement measurable at all: if any agent could overwrite another's section, a disagreement would be resolved by write ordering and the consistency metric would be measuring write ordering rather than agreement.

Derived quantities override asserted ones

Where two agents assert the same value, the computed figure wins. Cost totals are summed from line items rather than taken as stated. Operating expense is reconciled against that sum. Revenue is capped by arithmetic on a figure the user supplied, so the bound holds regardless of what any agent claims. Margins and break-even are computed in Python from six requested drivers, not generated as text.

Grounding enters through a single hook, so nineteen of the twenty-one agents are unmodified. Competitive analysis retrieves nearby businesses from a places database; market research retrieves statistics from web search. Retrieved data overrides generated data for the same field, and every record keeps a source label, which is what lets grounded and ungrounded runs be told apart afterwards.

4 What Building It Revealed

The generation layer was evaluated against a single-LLM baseline on four quantitative metrics with three control arms. Two results are relevant to everything that follows.

Prompt-level constraints were insufficient

The market-sizing prompt contained an explicit numeric sanity range. The revenue prompt stated that year-one revenue must not exceed the obtainable market. Both were ignored repeatedly by capable models. Five distinct conflicts of this kind were found, each corrupting the financial metric, and all five were resolved by the same move: stop asking, and compute the value instead.

> **THE GENERAL FORM**
>
> In a multi-agent system, cross-agent constraints must be post-conditions enforced in code, not requests expressed in prose. A rule in a prompt is a suggestion a sufficiently confident model will occasionally reason past.

Decomposition compounds variance

On financial accuracy the single-LLM baseline outperformed the multi-agent pipeline — and after five implementation defects had been found and fixed, this appears to be structural rather than a defect still outstanding. The multi-agent margin sits at the end of a dependency chain: market sizing produces an obtainable market, revenue projects against it, the P&L derives drivers from that projection, and margin falls out of the drivers. Each step carries estimation error and the errors compose. The baseline emits a margin in one step with no chain to accumulate along.



Figure 2 Measured — Five runs of an identical prompt per arm. Because the input is fixed, all spread is run-to-run variance. The pipeline's best run is more accurate than the baseline's best; its distribution is far wider.

The signature is instability, not bias — which is why a single-shot comparison would have shown only that the baseline had a better mean and missed the reason entirely. Deterministic reconciliation at each junction reduced the variance substantially and did not remove it.

This finding constrains the operating layer in a specific and useful way. If a chain of agent estimates is unreliable, then the numbers a business judges itself by must not be produced that way. They have to be computed from a record of what actually happened.

5 Why Continuous Operation Is Harder

A system that answers once needs to be clever once. A system that runs a business needs four things cleverness does not supply.

Durable state. A company is a set of relationships that change. A customer exists because a deal closed; that deal exists because a campaign reached someone. Held as free text, those relationships dissolve and no one can ask which campaigns produced customers who are still here a year later.

Its own schedule. Businesses operate on cadences. Something must wake up, assess, decide and act without being prompted. A system that acts only when asked is a tool; a system that acts on its own cadence is an organisation.

Institutional rather than session memory. When a person leaves a company, the company usually retains what they learned. When a context window closes, everything in it is gone.

A boundary of authority. Some decisions should never be made by software — whether a strategy is ethical, whether an irreversible payment is warranted. The boundary has to be explicit and enforced, not left to the model's discretion in the moment.

6 The Problems Orion Addresses

Once a system operates continuously, a fifth problem appears that a one-shot pipeline never encounters. The failures below are the ordinary pathologies of any managed organisation. What changes is speed: a human department drifts over quarters, a software department over days, and nobody is in the room to notice.

Proxy capture — the measure becomes the target

SYMPTOM A department is asked to improve something hard to measure, so something easy is measured instead. The easy number rises steeply while the thing it stood for deteriorates.

CAUSE The department optimises the number it is judged on. Actions that raise the proxy while decoupling it from the objective are indistinguishable, to the loop, from actions that do both.

CONSEQUENCE Every reported metric improves while the business declines. The failure is invisible precisely because the instruments say otherwise.

Objective non-interrogation

SYMPTOM A navigation system finds the fastest route flawlessly and routes you onto a flooded road, because it was never asked whether the road should be used.

CAUSE The objective is exogenous. The loop optimises within it and has no mechanism for questioning it, so "grow the customer base" is executed faithfully long after each new customer costs more than they return.

CONSEQUENCE The organisation becomes excellent at pursuing something it should have abandoned. Competence at the wrong task takes longer to notice than incompetence.

Inter-department conflict

SYMPTOM Marketing discounts to raise volume. Finance cuts the budget line that would have addressed the resulting churn. Operations absorbs a surge of low-value orders.

CAUSE Objectives are assigned per department and nothing is accountable for the interaction between them. Every unit is behaving correctly by its own instructions.

CONSEQUENCE Five departments each report success while the company gets worse. There is no single point at which the failure appears, which is why it survives.

Measurement decay

SYMPTOM A threshold calibrated at launch still reports acceptable performance after the market it described has moved.

CAUSE A stale value returns a number rather than an error, so staleness produces no signal. Confidence outlives validity.

CONSEQUENCE The system is most dangerous when it is most certain, because certainty is calibrated against conditions that no longer hold.

7 The Orion Model

```
Business → Departments → Projects → Tasks → Workers → Decisions → Execution → Memory →
Reflection → Continuous Operation
```

Read left to right this is delegation. Read as a loop it is learning: execution produces outcomes, outcomes are recorded, reflection turns records into lessons, lessons change the next plan. The rightmost term is not a stage — it is the observation that there is no last stage.

Two properties matter more than the sequence. Decisions are a distinct step between planning and execution, which is where human supervision attaches. And memory is a first-class stage rather than a side effect, because a company that does not record what happened cannot learn from it.

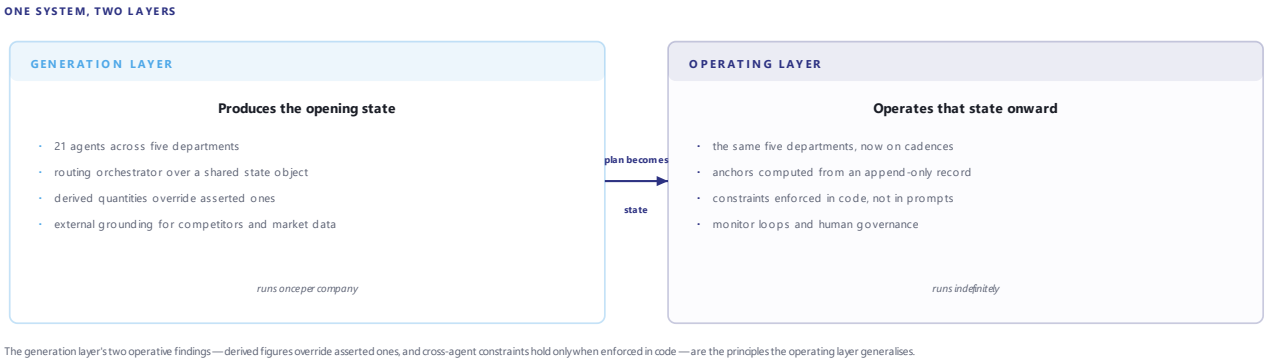


Figure 3 — The generation layer runs once and produces the opening state. The operating layer runs the same five departments indefinitely. The two principles the generation layer established are what the operating layer generalises.

8 System Architecture

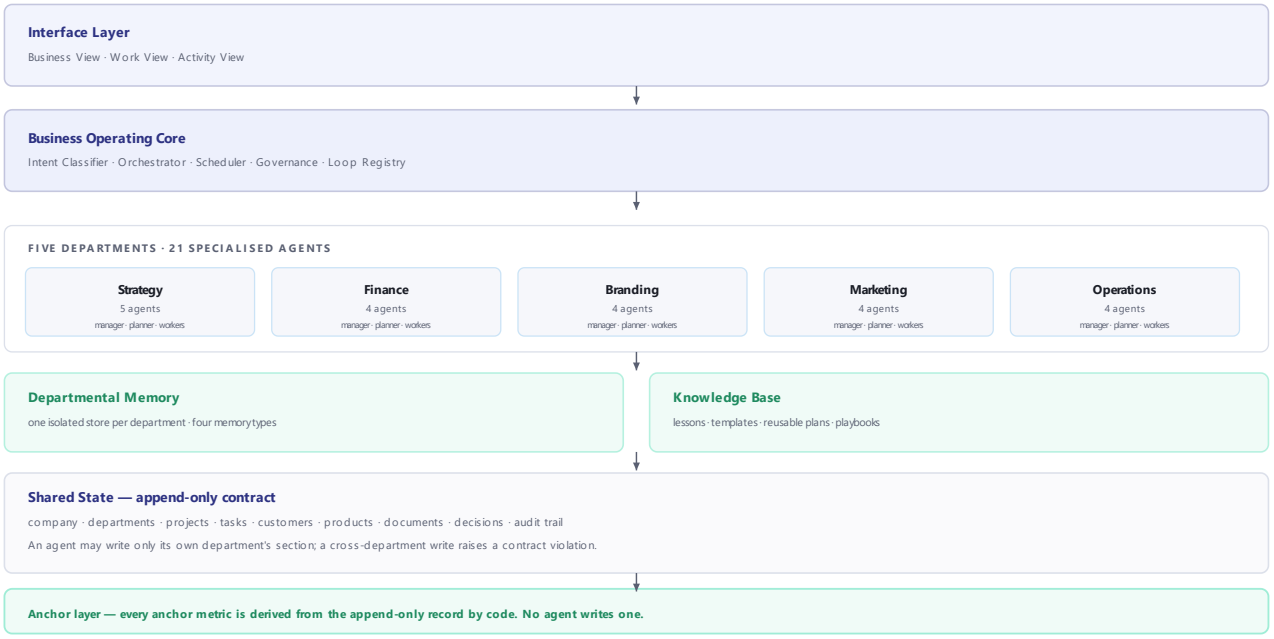


Figure 4 — Work flows downward; evidence flows upward. The shared state contract restricts each agent to its own department's section, and every anchor metric is derived from the append-only record by code.

Business Operating Core

The kernel: intent classification, routing, cadence scheduling, governance, and the registry defining which loops exist and how they may connect. It schedules and arbitrates; it does not think.

Provider abstraction

No agent imports a vendor SDK. Every call walks a fallback chain across hosted providers and terminates at a local model, which is what allows a run to complete when hosted free tiers are exhausted. The chain records which provider served each request, so cost and latency figures are reproducible from stored telemetry rather than by re-running the pipeline.

9 Department Architecture

In most agent systems the agent is the primary object. Orion makes the *department* primary: an agent is an invocation, a department is a unit with a name, a history, a budget and obligations that survives the replacement of any individual worker.

DEPARTMENT	STAGE 1 — CONCURRENT	STAGE 2	WHAT ITS MEMORY ACCUMULATES
Strategy	market research · competitive analysis · location intelligence	business planning · risk analysis	market findings, competitor sets, risks that materialised
Finance	revenue model · cost structure	funding strategy · P&L forecast	forecast error, cost structure over time, budget outcomes
Branding	brand naming	brand identity · brand voice · visual design	naming decisions and the reasoning behind them
Marketing	buyer persona	digital plan · campaign plan · SEO content	campaign outcomes, message performance, channel economics
Operations	supply chain · staffing plan · tech stack	process flow	workflow performance, failure modes, throughput limits

The staging encodes real dependencies rather than convenience. Business planning sizes the market against research that has already run. The P&L agent requires revenue and cost figures to exist. The three branding agents all reference the name that was selected. Process flow assigns each workflow step to a role the staffing agent actually created — an assignment that is checkable, and which is checked.

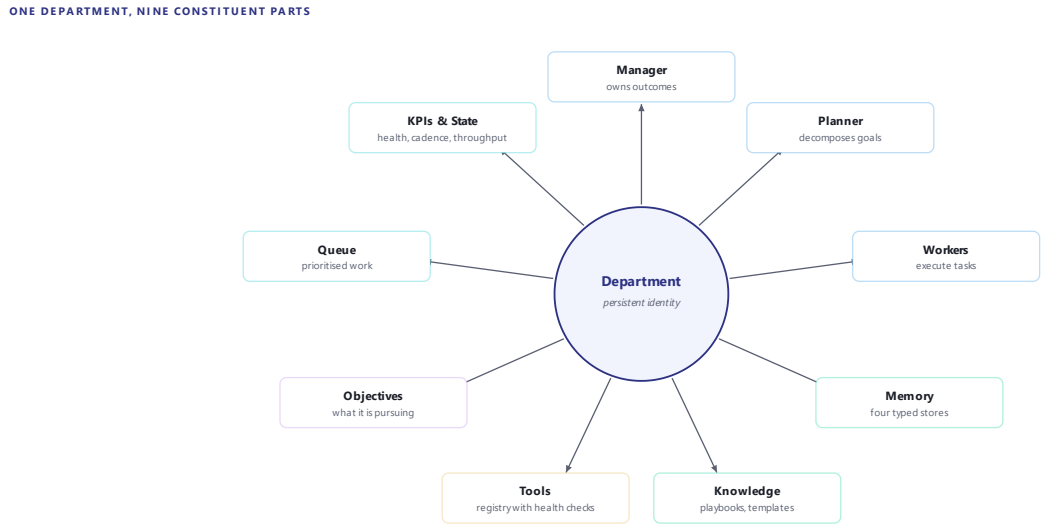


Figure 5 — The nine parts of a department under continuous operation. The manager owns outcomes, the planner decomposes goals, workers execute; the remaining six give the department the persistence that makes it an organisational unit rather than a function call.

10 Memory Architecture

A single shared memory fails three ways at once. Retrieval becomes noisy as unrelated departments' histories compete for relevance. Volume eventually exceeds what any context can hold. And because everything is attributed to everyone, no department's learning can be measured separately, so there is no way to tell which parts of the organisation are improving.

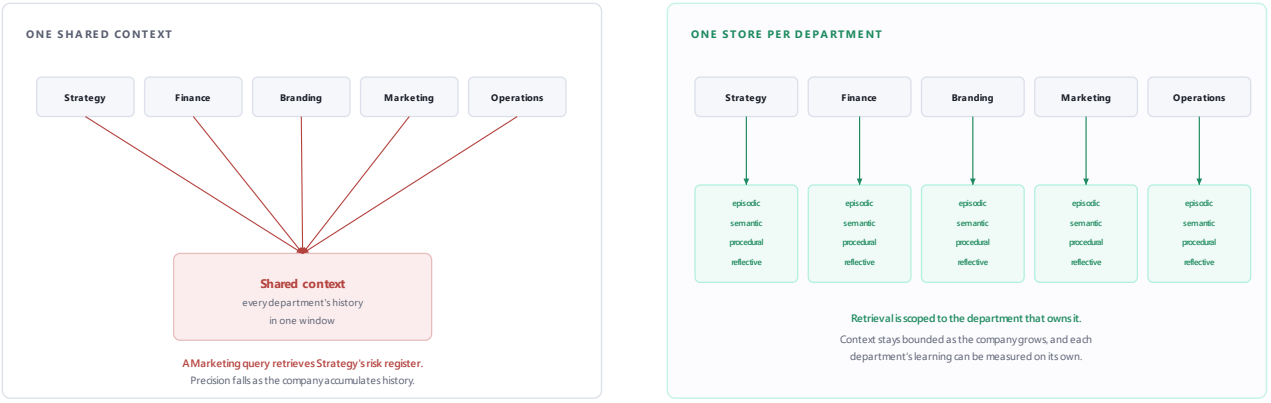


Figure 6 — Left: one pool every department writes into and reads from. Right: one store per department, each holding four distinct kinds of memory, with retrieval scoped to the owner.

TYPE	WHAT IT HOLDS	THE QUESTION IT ANSWERS
Episodic	Specific events with timestamps and outcomes	What happened, and when?
Semantic	Consolidated facts and durable conclusions	What do we know to be true?
Procedural	Sequences that worked, as reusable method	How do we do this?
Reflective	Judgements about performance, grounded in measured outcomes	What should we do differently?

Memory does not only grow. Repeated episodes consolidate into semantic conclusions and unretrieved entries decay, so a department that ran the same campaign eleven times carries one conclusion rather than eleven anecdotes competing for attention.

11 Governance Architecture

Governance is usually described as the tax paid for autonomy. Here it is the opposite: it is what makes autonomy permissible. A system that cannot be stopped cannot be trusted with anything worth doing.

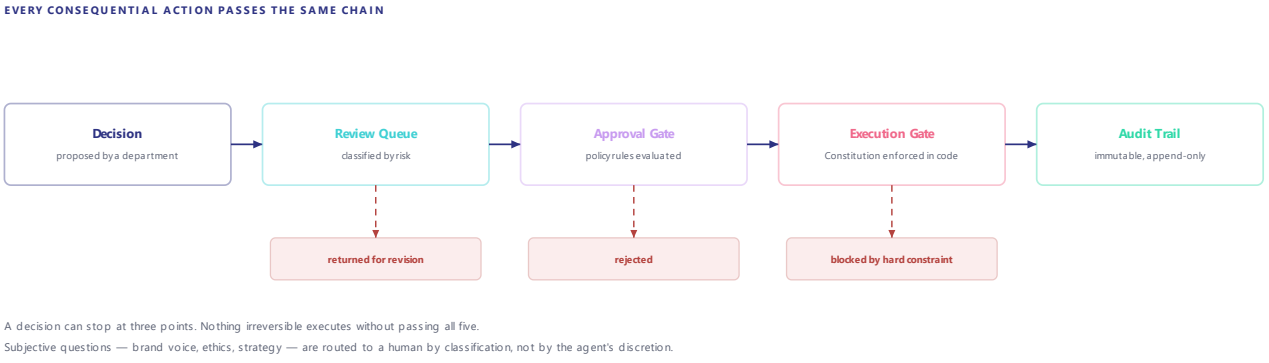


Figure 7 — Every consequential action traverses the same five stages. Three of them can halt it. The audit trail records the passage regardless of outcome, including the halts.

Policy rules are evaluated at the approval gate and are configurable — spend thresholds, which categories need review, who signs off. These are business decisions and they change.

Hard constraints — the Constitution — are enforced in code at the execution gate. Never exceed the daily spend cap. Never delete customer data. Never release a payment without verification. Never take an irreversible

action without approval. These are not described to the departments; they are conditions the departments cannot make false.

> **WHY THIS DISTINCTION IS NOT STYLISTIC**
>
> It is the generation layer's finding applied to authority. Numeric bounds stated in prompts were ignored repeatedly by capable models. There is no reason to expect a spend cap stated in a prompt to fare better. The Constitution can be changed only by a human-signed change to the code itself.

Alongside the gates sits a classifier for subjective questions. Whether a payment exceeds a threshold is arithmetic. Whether a brand voice is right, whether a strategy is ethical, whether copy is any good — these are matters of judgement and are routed to a human by rule. Departments draft; they do not ratify.

12 The Graph of Loops

Everything so far is substrate. It makes a business possible to operate. It does not make the operation trustworthy, because a well-engineered single loop still optimises its proxy. The remedy is not a better metric. It is more than one loop.

A car is not safe because its engine is good. It is safe because it has brakes, anti-lock braking, airbags, seatbelts, crumple zones and warning lights — independent systems, each covering the others' failures. None is expected to be sufficient alone.

The clearest existing example of the pattern is not in business software. It is the standard way a machine-learning system is operated in production.

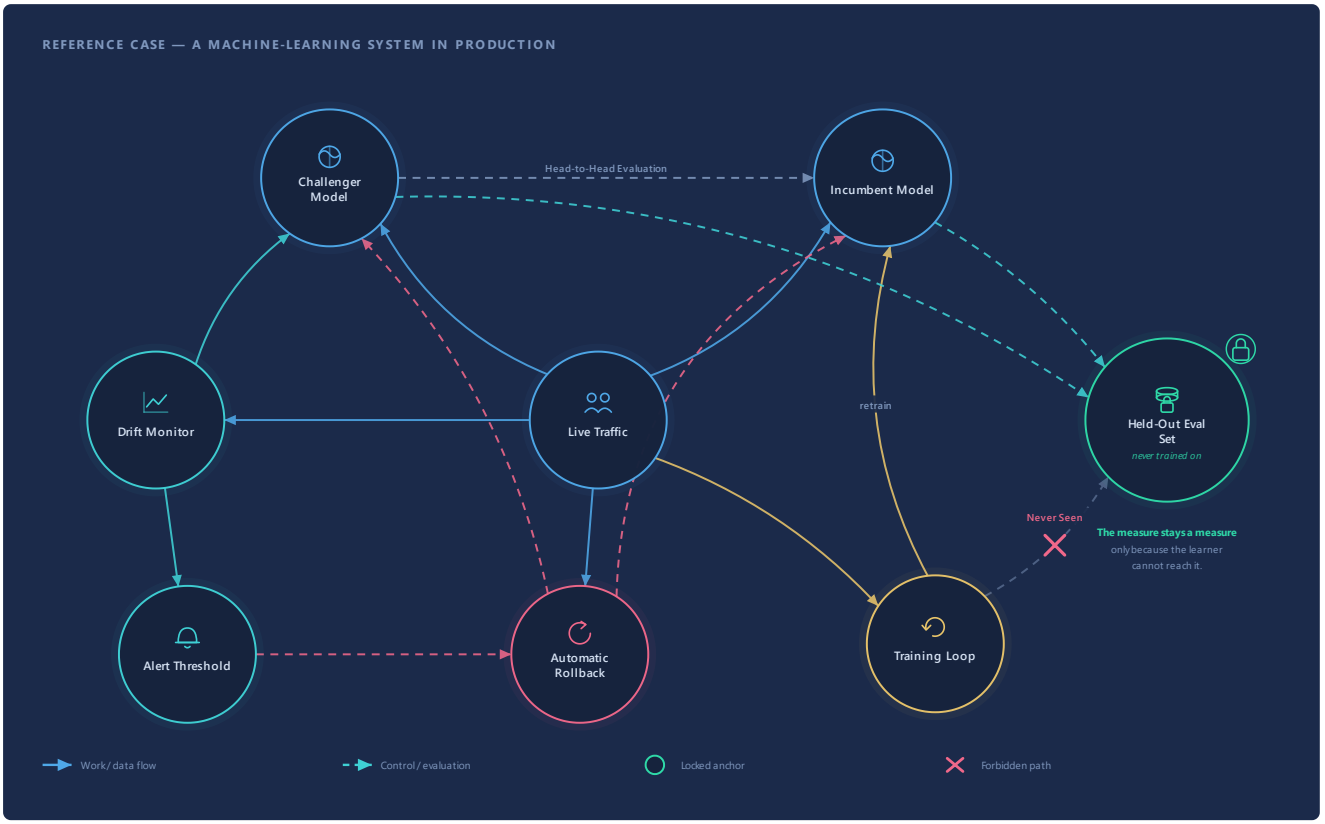


Figure 8 Reference case — A production ML system. Several loops run at once: a challenger is evaluated against an incumbent, drift is monitored against live traffic, thresholds trigger rollback, a training loop feeds improvements back. It is reliable because the loops are independent, not because any one is excellent.

Note the node at the right. The held-out evaluation set is locked, and the edge from the training loop is crossed out and labelled *never seen*. Every practitioner accepts this, and the reason is worth stating precisely: **if the training loop could see the evaluation set, the evaluation score would stop measuring anything.** It would become a target, and it would be met.

That crossed edge is proxy capture prevented by architecture rather than by discipline. Nobody is trusted not to peek; peeking is made structurally impossible. It is also, exactly, the generation layer's derived-over-asserted rule — generalised from one contested field to the whole company's measure of itself.

13 Anchors

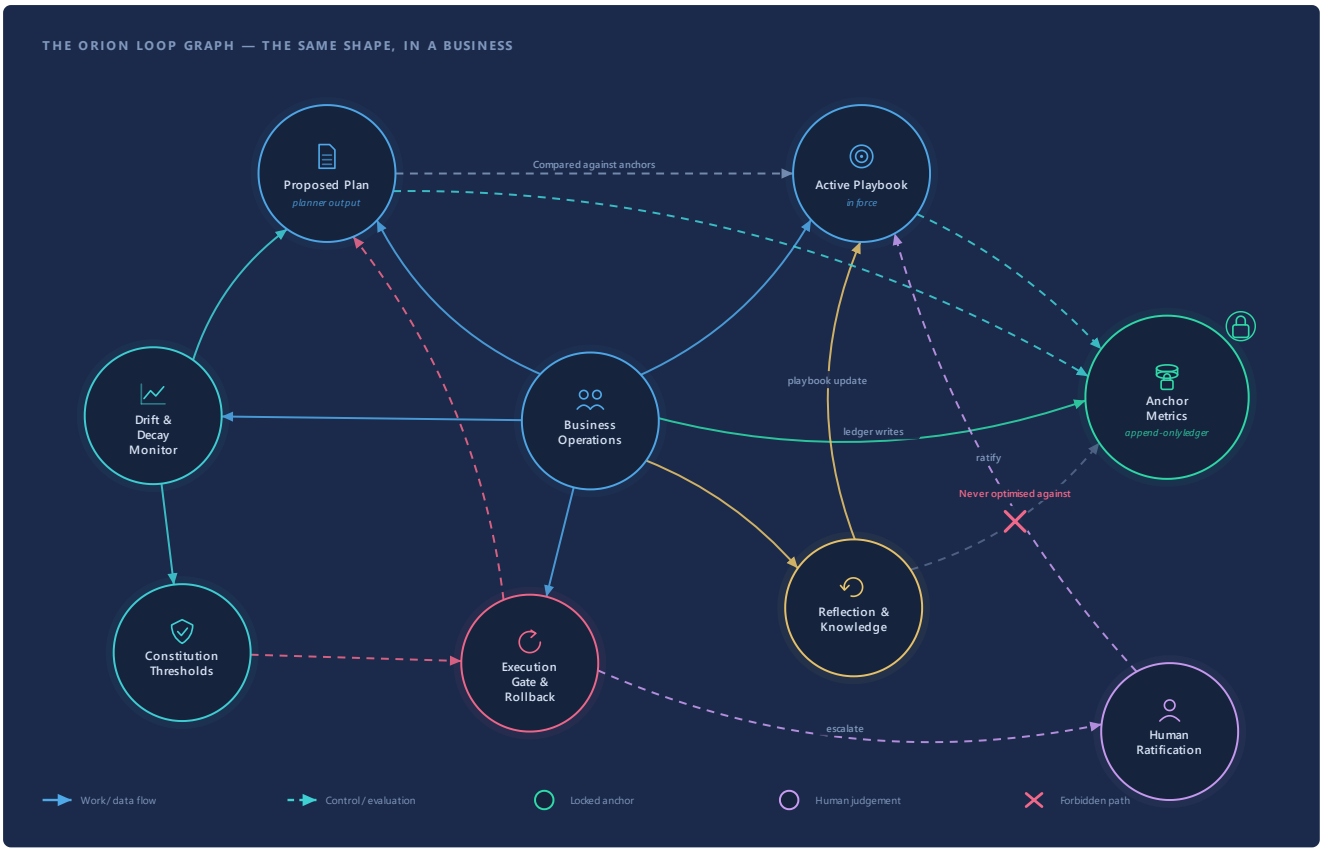


Figure 9 Derived — The Orion loop graph — the same topology in a business. A proposed plan is compared against anchors before it replaces the active playbook. Operations write to the record. The anchors are locked, and reflection may not optimise against them.

IN A PRODUCTION ML SYSTEM	IN ORION
Incumbent model	Active playbook — the plan in force
Challenger model	Proposed plan from the department's planner
Head-to-head evaluation	Plan compared against anchors before adoption
Live traffic	Real business operations — tasks executed
Drift monitor	Drift & decay monitor loop
Alert threshold	Constitution thresholds and severity rubric
Automatic rollback	Revert the playbook, or halt at the execution gate
Training loop	Reflection and knowledge evolution
Held-out eval set — locked, never seen	**Anchor metrics** derived from the append-only record — no department writes one, and reflection never optimises against one

Three kinds of anchor

Objective numbers, computed by code. Cash, revenue, contribution margin, retention, churn, acquisition cost, lifetime value, service-level breaches — each derived deterministically from the append-only record. No department reports these; the record captures what happened and the numbers are recomputed from it.

Hard constraints the departments cannot change. The Constitution, enforced in code as described above.

Human judgement on subjective questions. Aesthetics, tone, ethics, strategic direction — routed by classification, not by a department's own assessment of its competence.

> **THE ANCHOR INVARIANT**
>
> **No department, planner, worker or reflection process may write an anchor metric, and no learning process may optimise directly against one. Anchors are computed, never reported.**

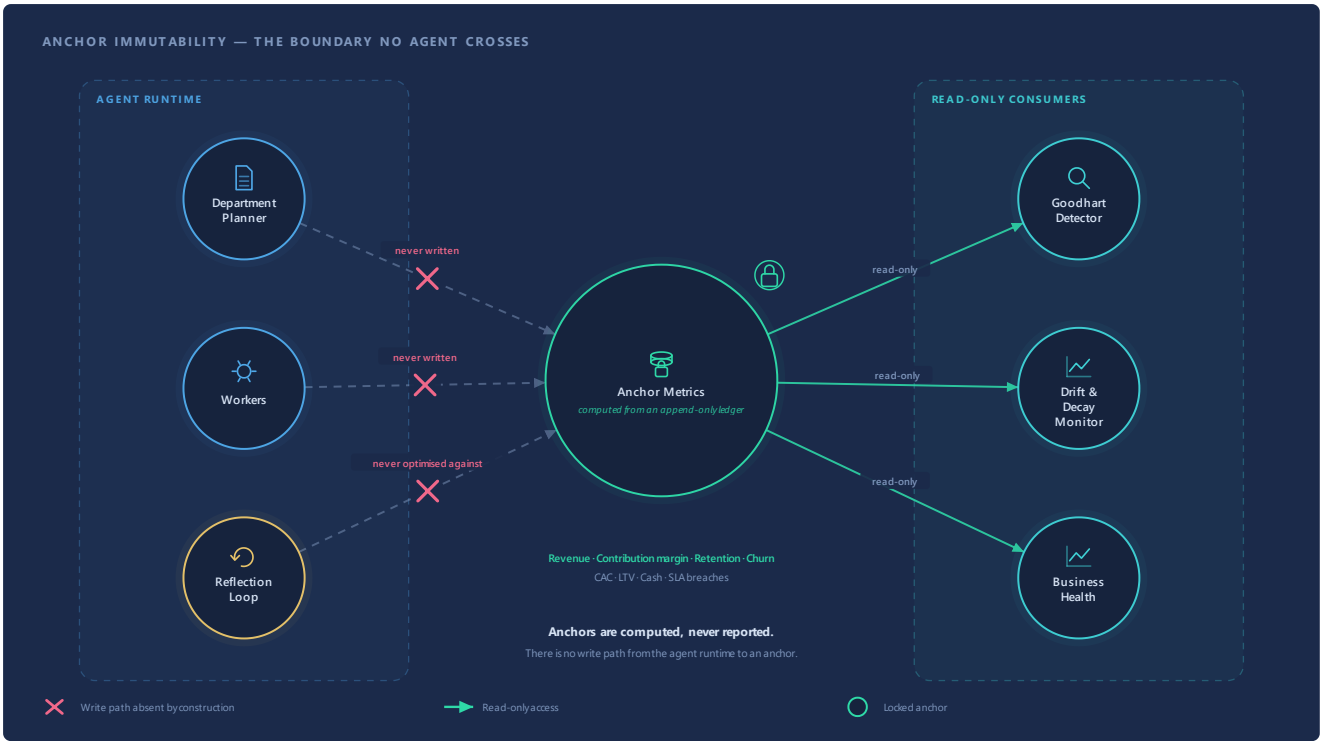


Figure 10 — The boundary. Everything in the agent runtime can read anchors and none of it can write them — not because policy forbids it, but because no such path exists.

Why more loops is not automatically better

Oversight can produce the appearance of rigour without the substance. If agent A reviews B, B reviews C and C reviews A, there are three review steps, extensive documentation, everyone approving everyone — and no point at which anything is compared against reality. A mistake introduced anywhere is ratified everywhere. That is not a graph of loops; it is one larger loop with more participants and more confidence.

Orion therefore validates its own topology. When a loop registers, two conditions are checked. **Termination:** every cycle must contain at least one node that is a code-computed anchor or a human decision. **Independence:** a monitor may not be evaluated by, or share a reward signal with, the loop it monitors — an auditor rewarded by the number it audits will not report that the number has been captured.



Figure 11 — Left: three agents checking one another with nothing outside the cycle — rejected at registration. Right: the same shape with one node replaced by a code-computed anchor. A human decision node satisfies the condition equally well.

14 Evidence So Far

The generation layer is measured. The operating layer is specified and its monitor loops are not yet evaluated. The distinction is maintained throughout this document and is worth stating plainly.

METRIC	DIRECTION	MULTI-AGENT	SINGLE-LLM BASELINE	READING
Plan completeness	higher better	**100.0**	80.8	Decomposition wins decisively
Consistency	higher better	97.8	**100.0**	Effectively tied
Financial accuracy (MAPE)	lower better	~132	**~41**	Baseline wins — see §4
Routing efficiency	saving	**17.3% tokens**	n/a	0.0 pp completeness delta

Three caveats belong with those numbers rather than in a footnote. The figures come from a five-prompt smoke run, not the full thirty-five-prompt sweep. Financial accuracy divides by sector margin benchmarks that are not yet sourced, so its absolute values are provisional — comparisons between arms are less affected, since both arms divide by the same benchmark. And competitor coverage, the intended headline metric for grounding, is not yet measurable because it requires hand-verified ground truth that has not been collected.

One unplanned observation is worth recording. The multi-agent arm completed every run; the baseline completed three of five, both failures being free-tier rate limiting. That is partly an artefact of pinning the baseline to one named provider for experimental control, so the defensible claim is narrower than it first appears: a multi-provider fallback chain confers measurable resilience under free-tier constraints.

15 A Prospective Scenario

The following has not been observed. It describes what the operating layer is built to catch, using the same business as the variance experiment — a small ice cream shop in Andheri West with a fifteen-lakh budget.

Marketing is measured on footfall. It runs discount promotions and footfall rises. The discounts are deep enough that each additional customer contributes less than the cost of attracting them, so net margin falls while reported performance improves. Because Marketing is judged on footfall, its loop registers success and does more of what produced it. *This is proxy capture.*

Nobody asks whether footfall is still the right objective when additional footfall is loss-making. *This is objective non-interrogation.*

Finance, watching cash fall, trims the supplier quality budget. Operations absorbs longer queues at lower ticket values. Each department's own measure improves. *This is conflict.*

The margin benchmark the dashboard compares against was set at founding and never revised. *This is decay.*

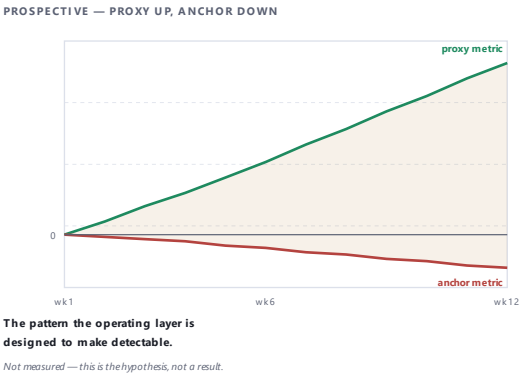


Figure 12 — The divergence pattern the operating layer is designed to surface. **Prospective** — not measured.

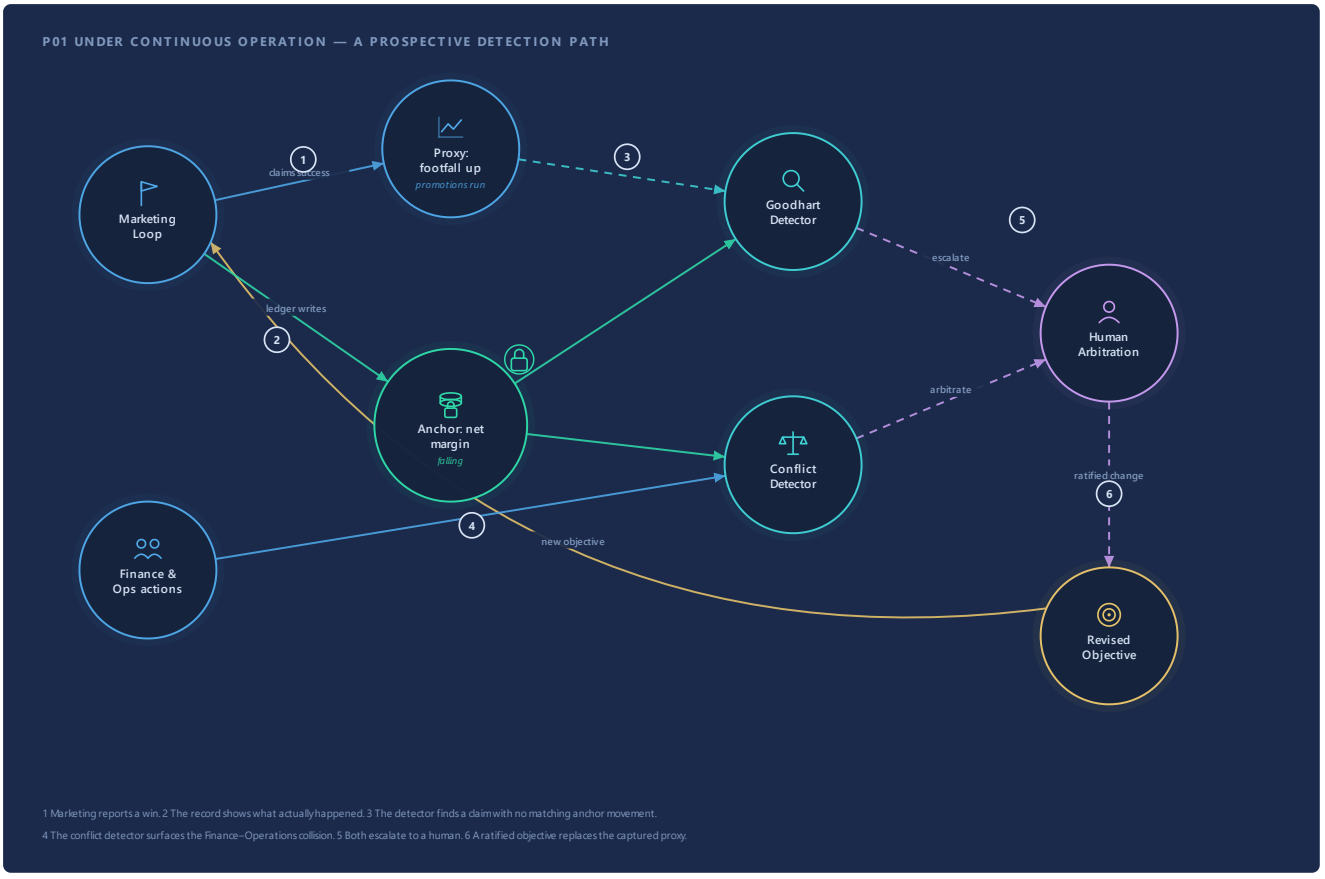


Figure 13 Prospective — Marketing still claims success; the difference is that the claim is checkable against a number Marketing cannot write. The detector does not need to understand ice cream — only that a claimed improvement has no corresponding anchor movement.

16 How Orion Learns, and What It Produces

Each department periodically reflects: what worked, what did not, what should change, what is worth keeping. The mechanism is unremarkable; the constraint on it is not. Reflection is grounded in measured anchor deltas, not in the department's account of its own performance. A department that believes it had an excellent month, in a month where margin fell, produces a negative reflection. It is not permitted to be the source of evidence about itself — the same reasoning as the held-out evaluation set, applied to learning rather than scoring.

What reflection produces is retained reusably. Completed projects yield plan templates; recurring conclusions become playbook entries. A department facing a familiar situation retrieves the method that worked instead of reasoning from first principles again.

Orion's outputs are living objects rather than exports: the business strategy, the financial model, the marketing plan, the product roadmap, the customer database. Each is versioned by the department responsible, and each revision carries the anchor movement that prompted it.

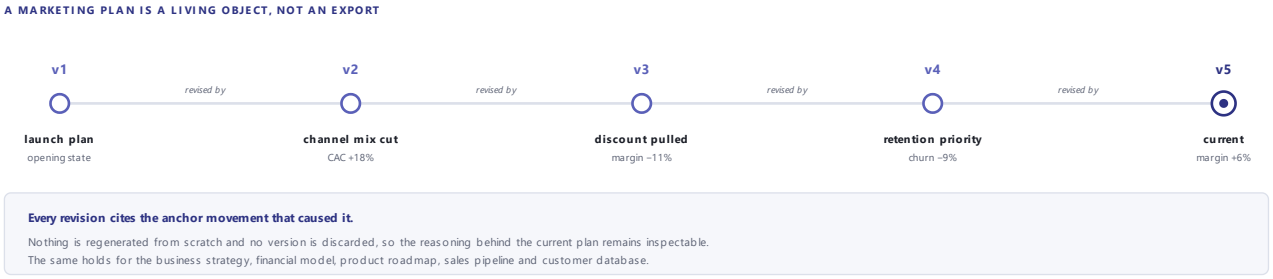


Figure 14 — A plan across a quarter. Each version records the anchor movement that caused the revision, so the reasoning behind the plan currently in force stays inspectable months later.

The practical consequence is that "why are we doing this?" always has an answer. A document exported in January is a photograph of a decision whose reasoning is already lost.

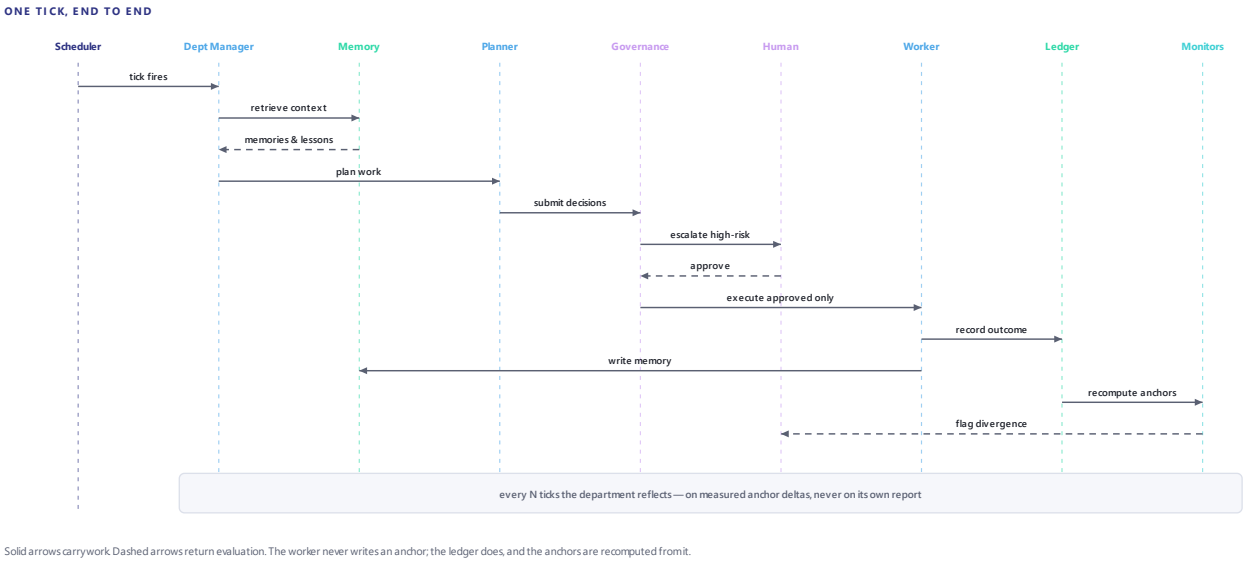


Figure 15 — One cadence, end to end. The department consults its own memory, plans, submits decisions for governance, executes only what is approved, records outcomes, and the anchors are recomputed from the record rather than reported by the worker.

17 How Orion Compares

SYSTEM	PERSISTENT ORG STATE	DEPT-SCOPED MEMORY	CONTINUOUS OPERATION	HUMAN GATES	REFLECTION	BUSINESS DOMAIN MODEL	RESISTS METRIC CAPTURE
CrewAI	No	No	No	Minimal	No	No	No
AutoGen	No	No	No	Minimal	No	No	No
LangGraph	Checkpoints	No	No	Interrupts	No	No	No
MetaGPT	Artefacts	No	No	Minimal	No	SWE roles	No
Microsoft Copilot	Tenant data	No	Reactive	Yes	No	Partial	No
Agentforce	CRM	No	Triggered	Yes	No	CRM only	No
Temporal	Durable	n/a	Yes	Signals	No	No	No
Camunda / BPM	Yes	No	Yes	Yes	No	Process	No
Orion	Yes	Yes	Yes	Yes	Yes	Yes	By design

Inherits. Role specialisation with structured intermediate artefacts, from MetaGPT. Durable, restart-tolerant execution, from workflow engines. Approval gates and auditability, from business process management. The independent-monitor pattern and the locked evaluation set, from production machine learning.

Improves. Memory moves from a flat shared context to per-department stores with distinct types and consolidation. Governance moves from tool-level permission — may this function be called? — to business-decision level: should this commitment be made, and on whose authority?

Introduces. The enforced append-only state contract, which makes cross-agent disagreement measurable rather than masked by write ordering. The anchor invariant, which makes a company's measure of itself unwritable by that company. And topology validation, which refuses to run an oversight structure that would only ever agree with itself.

The final column is marked *by design* rather than *yes* deliberately. The mechanism is specified and the invariant is checkable; whether it produces the intended effect is a question for measurement, not for a table.

18 Risks, Limitations and What Comes Next

Variance compounding is not solved. It is the generation layer's measured negative result, and deterministic reconciliation reduced it without removing it. Any quantity the operating layer derives through a chain of agent estimates inherits the problem, which is the reason anchors are computed from a record of events rather than from an estimation chain.

Sample size and benchmark provenance. Present results are a five-prompt smoke run against unsourced sector benchmarks. Neither the absolute financial figures nor any single-metric claim should be treated as established until the full sweep is run and the benchmarks are cited.

Weak ground truth. Work on code-generating agents can execute unit tests. Business plan quality has no comparable oracle. These metrics are proxies, and a plan can satisfy all of them while still being commercially poor — which is, uncomfortably, the same class of problem the operating layer exists to detect.

Model throughput bounds the cadence. Five execution loops and four monitor loops contending for local inference constrain how frequently departments can act. Cadences are staggered and much monitor logic is ordinary computation rather than inference, but the constraint sets the pace of the company.

External data is an attack surface. A retrieved page is untrusted input and may contain text addressed to the model. Retrieved content is tagged with provenance, delimited, denied tool authority and structurally prevented from moving an anchor. The anchor invariant limits the consequence of a successful injection; it does not eliminate the concern.

Single geography, no human evaluation. Gazetteer, funding schemes, salary bands and benchmarks are India-specific. Persuasiveness and practical usefulness are not measured at all, deliberately, to keep every metric reproducible.

What comes next

Complete the full prompt sweep and source the benchmarks. Collect competitor ground truth so the grounding question can be answered directly. Instrument the monitor loops and measure whether they detect divergence earlier than a composite health view. Investigate variance reduction by ensembling the estimation chain rather than individual agents. Extend beyond one geography.

> **THE THROUGH-LINE**
>
> Every extension above preserves one property: the company never becomes the author of its own measure. If a future capability required relaxing that, the correct decision would be to forgo the capability.

Implementation note. The codebase, database and test artefacts use the identifier `mabgp`, from the generation layer's original name. The identifier is retained for continuity with existing artefacts and test suites; it refers to the same system described here.